

El funcional de evaluación como elemento del bidual

Ejemplo 1 (Funcional de evaluación). Sea $x \in X \setminus \{0\}$ espacio normado. Definimos

$$\begin{aligned} \Lambda_x: X^* &\longrightarrow \mathbb{K} \\ L &\longmapsto \Lambda_x(L) = L(x) \end{aligned} \implies \|\Lambda_x\| = \sup_{\|L\|=1} |L(x)| \leq \sup_{\|L\|=1} \|L\| \|x\| = \|x\|.$$

Por el teo-esp-normado-separacion-punto-cero/Teorema 1, $\exists L_0 \in X^* : \|L_0\| = 1 \wedge L_0(x) = \|x\|$, luego $\|\Lambda_x\| \geq |L_0(x)| = \|x\|$. Es decir, $\|\Lambda_x\| = \|x\|$ y $\Lambda_x \in X^{**}$.

Referenciado en

- teo-isometria-biyectiva-reflexividad
- teo-esp-reflexivo-iff-dual-reflexivo
- esp-normado-reflexivo
- lem-acotado-dual-imp-acotado

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